Step 3: Design Deep Dive

After finalizing the high-level architecture, the next step is to **deeply analyze critical components of the system**.

What Should Already Be Done Before Step 3

At this stage, you should already have:

```None
✔ Clear understanding of requirements and scope

✔ High-level system architecture (Step 2)

✔ Feedback from interviewer on design

✔ Identified key areas for deep dive
```

Core Goal of Step 3

In this step, you:

- Identify **important system components**
- Deep dive into **critical design areas**
- Discuss **trade-offs, bottlenecks, and optimizations**
- Show your ability to design **real scalable systems**

```None
Goal = Go from high-level architecture → detailed component design
```

Key Mindset

- Work with interviewer like a teammate
- Prioritize **important components only**
- Avoid unnecessary micro-details
- Focus on **scalability + trade-offs**

```None
Depth > Breadth (but only in critical areas)
```

How to Approach Step 3

1. Identify Key Components to Deep Dive

Work with interviewer to decide focus areas such as:

- Database design
- Cache strategy
- API design
- Message queues
- Feed generation logic
- Search system
- Consistency handling

```None
Select components that impact performance and scalability most
```

2. Prioritize Components

Not all components deserve equal depth.

Focus on:

- Bottlenecks
- High traffic paths
- Complex logic systems

```None
Prioritize = High traffic + High complexity + Critical path
```

3. Time Management (Very Important)

Avoid going too deep into irrelevant topics.

Example of over-detailing:

- Deep diving into Facebook EdgeRank algorithm in detail

Why it's bad:

- Not needed for system scalability design
- Consumes interview time
- Doesn’t show core system design skills

```None
Avoid micro-optimizations unless explicitly asked
```

4. Show System Thinking

For each component, discuss:

- Why this design?
- What are alternatives?
- Trade-offs (latency vs consistency, cost vs performance)

Example: URL Shortener Deep Dive

Focus area:

- Hash function design
- Collision handling
- Encoding strategy

Key discussion points:

- Base62 encoding
- Uniqueness guarantees
- Scalability of key generation

Example: Chat System Deep Dive

Focus area:

- Real-time messaging
- Online/offline status

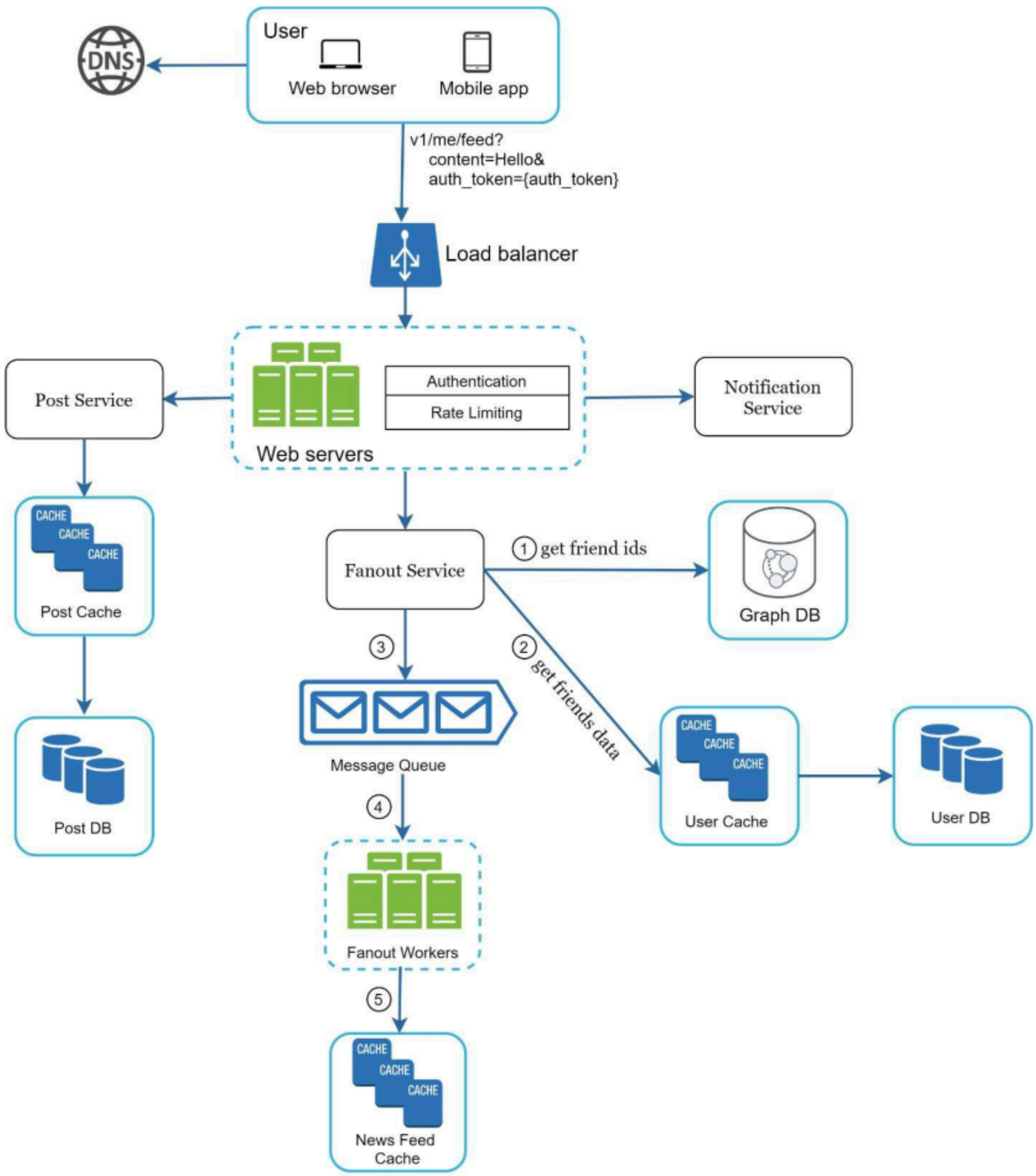
Key discussion points:

- WebSockets vs polling
- Message delivery guarantees
- Presence service design
- Reducing latency in message delivery

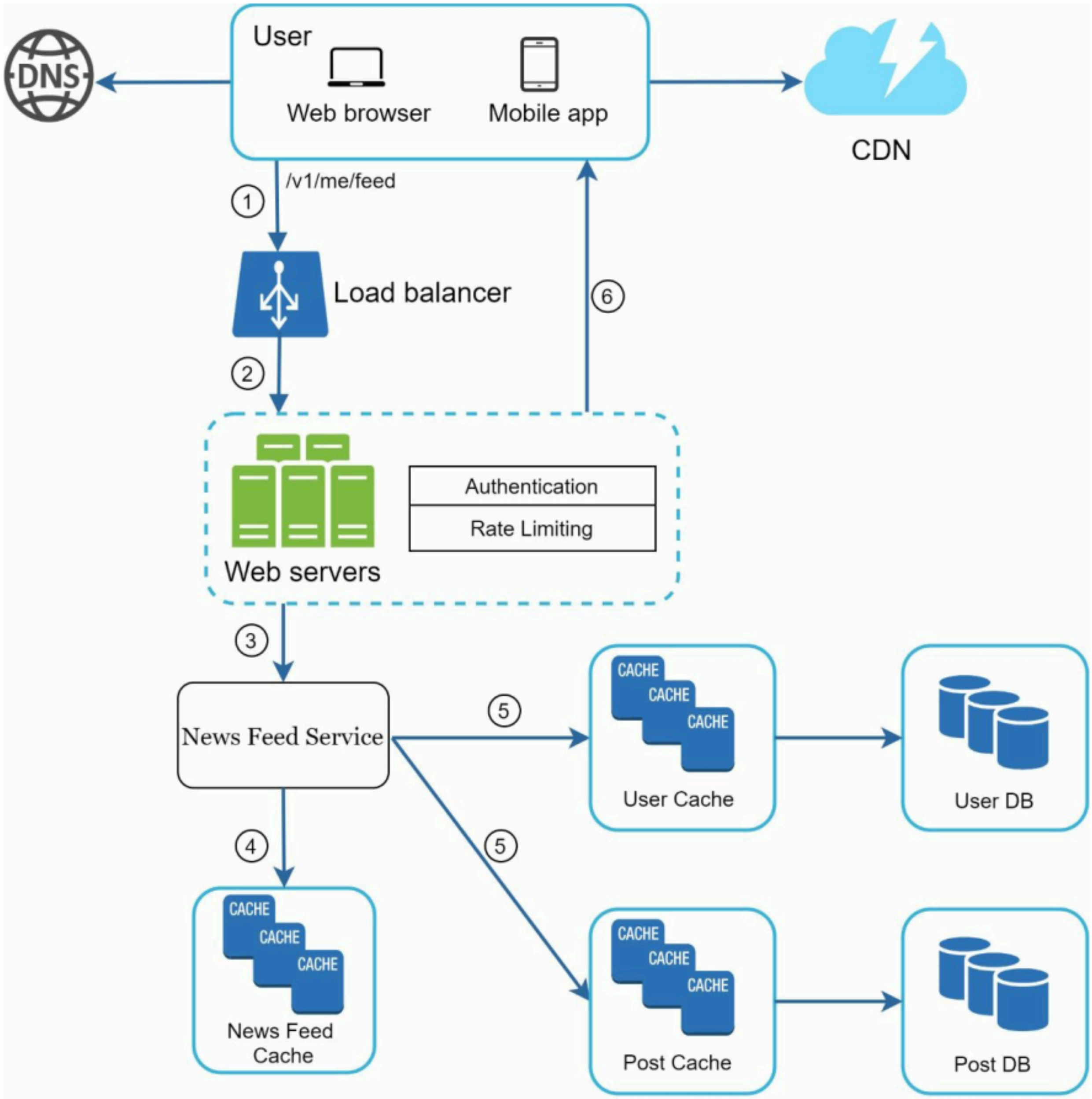
News Feed System Example

At this stage:

Feed Publishing Deep Dive



Feed Retrieval Deep Dive



Key Insight

```None
Step 3 is about identifying critical system bottlenecks and designing detailed solutions only for those areas.
```

Mental Model

```None
High-Level Design → Identify critical components → Deep dive → Optimize → Discuss trade-offs
```

One-Line Summary

Step 3 focuses on deeply designing only the most important components of the system, analyzing trade-offs, bottlenecks, and optimizations while avoiding unnecessary low-level details.